

Kernel Trick in SVM

1 The Problem with Linear SVM

- Linear SVMs work well **only if data is linearly separable** in the original feature space.
 - But in real-world data (like circles, spirals, XOR problems), data **is not linearly separable**.
 - **Example:** Imagine points forming two concentric circles. A straight line cannot separate them.
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2 Idea: Map Data to Higher Dimensions

- If data is not linearly separable in **2D**, maybe it becomes separable in **higher dimensions**.
 - **Example:**
 - In 2D: data looks like circles → not separable.
 - In 3D: lift it with a new feature $z = x^2 + y^2$, and suddenly you can separate them with a plane!
 - So, the idea is: **map input space → higher-dimensional feature space** where a linear separator exists.
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3 The Problem: High-Dimensional Mapping is Expensive

- Explicitly computing transformations like $x^2, x^3, \dots$ is computationally expensive.
 - For very high dimensions (or infinite), it's practically impossible.
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4 The Kernel Trick

Instead of explicitly mapping, we use a **Kernel Function** that directly computes the **dot product in higher dimensions**.

$$K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$$

- Here, $\phi(x)$ = feature mapping function (possibly infinite dimensional).
- Kernel function computes similarity in high-dimensional space **without ever computing the mapping**.

That's the trick: we **never leave the input space**, but still behave as if we did.

5 Common Kernel Functions

1. Linear Kernel

$$K(x_i, x_j) = x_i \cdot x_j$$

👉 No transformation, just standard dot product.

2. Polynomial Kernel

$$K(x_i, x_j) = (x_i \cdot x_j + c)^d$$

👉 Allows curved decision boundaries of degree d .

3. RBF (Gaussian) Kernel

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$$

👉 Projects to infinite-dimensional space; very powerful, handles complex boundaries.

4. Sigmoid Kernel

$$K(x_i, x_j) = \tanh(\alpha x_i \cdot x_j + c)$$

👉 Similar to neural networks' activation.



6 Why Kernel Trick Works in SVM

- In SVM dual formulation, only **dot products** of data points appear.
- So, if we replace dot products with kernel functions, we automatically work in higher-dimensional space **without extra computation**.

7 Visual Intuition

- In low dimensions → boundary is a straight line.
- With kernel → boundary can be a **circle, curve, spiral**, depending on kernel choice.
- RBF Kernel is the most common → makes flexible, smooth boundaries.

➤ Quick Recap

- Kernel trick allows SVMs to solve **non-linear problems**.
- It avoids the computational cost of explicitly transforming data.

- Different kernels = different ways of seeing similarity.
 - That's why SVM is powerful in both **linear and non-linear tasks**.
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